

Kalpiti Swami

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SUMMARY

Cloud and DevOps Engineer with experience automating infrastructure, CI/CD pipelines, and operational workflows on AWS. Built a GitOps-based EKS platform with zero static credentials using OIDC federation, Trivy scanning, and Prometheus/Grafana observability. AWS Certified Solutions Architect and 2x AWS JAM Podium.

EDUCATION

Seneca Polytechnic, Postgraduate Certificates: Cloud Architecture & Cybersecurity Sept. 2024 to Apr. 2026 | 4.0 GPA
Coursework: IT infrastructure, AWS, Microsoft Azure, Terraform, Ansible, Kubernetes, networking (TCP/IP, DNS), scripting, security principles.

EXPERIENCE

Information Technology Analyst, DevOps & Infrastructure (Co-op)

Jan. 2026 to Apr. 2026

Ministry of Health (MPBSDP), Government of Ontario

North York, ON

- Reduced Linux/Ubuntu OS provisioning time by **80% across 100+ VMs** at 21 sites by engineering a parallel-safe Ansible playbook with per-VM OVA templates and cloud-init, cutting per-batch runtime from **50 minutes to 5** in a hybrid cloud environment.
- Authored PowerShell automation for a **150-machine** Citrix VDA upgrade, compressing a **187-hour** manual process to **12 hours** of unattended runtime and eliminating an estimated **4 engineers' worth** of sequential manual effort.
- Coordinated **100+** infrastructure change requests spanning network patching, Windows patching, and data centre maintenance across **21 sites** through CAB approval in the eSMT portal, achieving a **100% approval rate** with **zero rejections** including several expedited changes.
- Automated device and circuit provisioning in **NetBox using Python** to update router records and create switch entries across **production network infrastructure**, with full documentation maintained for ongoing operational use.

PROJECTS

Fully Automated EKS GitOps Platform - Zero Static Credentials | [Infra GitHub](#) | [WebApp GitHub](#) | [Dev.to Blog](#)

- Hardened **CI/CD supply chain security** by pinning GitHub Actions to **immutable commit SHAs**, enforcing **Trivy container scanning** and **Checkov IaC validation** on every pull request, and blocking **CRITICAL severity findings** before image publication to ECR.
- Eliminated **long-lived AWS credentials** across CI/CD and runtime by implementing **GitHub Actions OIDC federation**, per-workload **IRSA scoped roles**, and **External Secrets Operator** syncing secrets dynamically from Secrets Manager.
- Automated full EKS environment standup from **zero to live in under 20 minutes** by engineering a sequenced pipeline covering Terraform provisioning, **Trivy-scanned ECR builds**, **ArgoCD Helm bootstrap**, and deployment validation, eliminating **all manual intervention**.
- Resolved **VPC deletion deadlocks** caused by Kubernetes-managed AWS resources outside Terraform state by implementing **ArgoCD cascade deletion finalizers** and layered destroy sequencing, achieving **fully automated environment teardown** with zero manual cleanup.
- Deployed **full-stack observability** on EKS by provisioning **kube-prometheus-stack**, **Grafana dashboards** via ArgoCD GitOps, and **8 CloudWatch Alarms** with SNS notifications, providing real-time visibility into cluster health, pod metrics, and infrastructure alerts.

Production-Style 3-Tier AWS Infrastructure with Terraform & CI/CD | [GitHub](#) | [Medium Pt.1](#) | [Medium Pt.2](#)

- Diagnosed and resolved **CloudWatch Agent failure** on ASG-launched EC2s by identifying **IMDSv1 incompatibility** and replacing metadata calls with **IMDSv2 token-based authentication** in user-data.sh, restoring full observability across all instances.
- Enforced **tier-to-tier traffic isolation** by implementing layered **security group chaining** across ALB, ASG, and RDS layers, eliminating direct public exposure of application and database tiers.
- Built **modular Terraform infrastructure** with remote state management and a **GitHub Actions CI/CD pipeline** gating all changes behind **manual approval on merge**, following GitOps-style deployment workflows.

TECHNICAL SKILLS

Cloud Platforms: AWS (EKS, EC2, ECR, VPC, IAM, Secrets Manager, ALB, S3, CloudWatch, SNS, VPC Interface Endpoints), Microsoft Azure (coursework)

DevOps and CI/CD: GitHub Actions, OIDC federation, Terraform, ArgoCD, Helm, Checkov, Trivy, GitOps

Containers and Orchestration: Docker, Kubernetes, EKS, HPA, ALB Ingress, EBS CSI Driver, RBAC, PersistentVolumes, StorageClass

Security: IRSA, External Secrets Operator, AWS Secrets Manager, zero static credentials, IMDSv2, least privilege, encryption at rest and in transit

Scripting and Version Control: Bash, Python (familiar), Git, GitHub, Azure DevOps

Observability: Prometheus, Grafana, Alertmanager, kube-prometheus-stack, CloudWatch dashboards and alarms, SNS notifications

Networking and OS: TCP/IP, DNS, HTTP, VPC subnets, NAT Gateway, load balancing, Linux (Ubuntu, Amazon Linux)

CERTIFICATIONS & AWARDS

- [AWS Certified Solutions Architect Associate](#) (SAA C03), 2026
- [AWS JAM](#): **1st Place** AWS Summit 2025, **2nd Place** AWS User Group 2026
- **4x President's Honour List**, 4.0 GPA across all semesters